

Software Technical Description

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Module: Architecture& Operating System
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The WAN architecture design features a well-sourced combination of proprietary and open-source software to fulfill the needs for performance, reliability, security, and scalability.

Software Component	Name & Version	License Type	Role / Purpose
OS (Client)	Windows 10/11 Pro	Proprietary	Client workstations
OS (Servers)	Windows Server 2019, Ubuntu Server 22.04 LTS	Proprietary (Windows), Open Source (Ubuntu)	Server OS for various roles
Server Roles	Active Directory, DNS, DHCP (Windows)	Proprietary	Network services for authentication, addressing
	Web Server (Apache), Mail Server (Ubuntu)	Open Source	Hosting web and email services
Database Management System	MySQL (primary), PostgreSQL (optional)	Open Source	Structured data storage and management
Backup Software	Ubuntu-based Storage & Monitoring tools	Open Source	Data backup and system monitoring

Configuration Management	Scripts & tools (Windows/Ubuntu)	Mixed	Automating config deployments and backups
Antivirus (Clients)	Windows Defender, Malwarebytes	Proprietary/Freemium	Endpoint protection on client PCs
Antivirus (Servers)	Defender for Endpoint	Proprietary	Enterprise-level protection for servers
Cloud Storage	Sync.com	Commercial	Encrypted cloud backup and file sharing

Operating Systems

For the clients, Windows 10/11 Pro was selected. They are common operating systems used within enterprises for their ease of use, high compatibility with software programs, and robust security features built into them *(Microsoft, 2018; Microsoft, 2021)*. The operating systems also accommodate enterprise-level device management features like Group Policy and BitLocker encryption, which assist in securing endpoints and implementing organizational IT policy.

The architecture on the server side utilizes both Windows Server 2019 and Ubuntu Server 22.04 LTS to take advantage of what each has to offer *(Microsoft, 2019; Canonical, 2022)*. Windows Server 2019 is used to host critical infrastructure roles such as Active Directory (AD), Domain Name System (DNS), and Dynamic Host Configuration Protocol (DHCP) services. All three are important for user authentication, managing domains, and allocating IP addresses in the network *(Microsoft, 2025)*, and for enabling a secure, central, and reliable way to control and provide consistent network activity.

Ubuntu Server, a popular Linux distribution, is used for running application services like web and mail servers, as well as backup and monitoring software. Ubuntu Server is preferred for its stability, security, and flexibility *(Canonical Ltd., 2025)*. Its open source makes it easy to customize and integrate with other software, which makes a difference when dealing with big data loads and assisting automated operational tasks. Having two distinct operating systems on the server maximizes the strength of both Windows' enterprise features and Linux's flexibility.

Server Roles and Services

WAN architecture keeps unstructured data services separate from structured application services to provide security and increased performance. Unstructured services such as Active Directory, DNS, and DHCP are installed on Windows Server to be used for the provision of basic network administration features *(Microsoft, 2025)*. They provide authentication requests, domain control, and assignment of IPs, which enable users and devices to connect and securely interact within the network.

Structured services, though, like mail and web server, are built on top of Ubuntu Server. The Apache HTTP server software operates the web server, which serves websites and web applications, and the mail server processes email messages using Postfix and Dovecot programs. Separating them keeps basic network administration activities isolated from application workloads, reducing the likelihood of security breach affecting basic network operations.

Backup and monitoring operations are also administered on Ubuntu servers using an open-source set of tools. Backup measures ensure data is stored securely at set times and can be restored when data is lost or when there is hardware failure. Monitoring programs such as Nagios or Zabbix can continuously monitor server and network device health, sending notifications when there's decreasing performance or system errors. This system, being proactive, ensures high availability and reduces downtime.

Database Management Systems

The WAN relies on MySQL 8.0 as its DBMS for storing structured data on the network *(Oracle,2024)*. MySQL is an open-source, robust

relational DBMS that enjoys widespread usage for its stability, scalability, and large support community *(Oracle Corporation, 2025)*. It effectively serves the data needs of web applications, mail systems, and monitoring databases.

In certain instances, PostgreSQL is used as an alternative or complementary DBMS. With its feature-rich nature, including support for complex query, JSON data types, and geospatial data types, PostgreSQL has flexibility for data-intensive applications that require advanced data processing *(PostgreSQL Global Development Group, 2024)* . The use of MySQL and PostgreSQL allows the WAN to support varied data storage requirements and facilitate best-in-class performance for different workloads.

Security Software

Security is an important aspect of the design for the WAN. On the clients, multi-layer protection against cyber security threats, viruses, and malicious software is delivered through a combination of Windows Defender and *Malwarebytes (Window, 2024; Malwarebytes,2024)* . Windows Defender gives real-time detection for threats and is nicely integrated into Windows OS features, while Malwarebytes adds another scanning engine for detecting specifically malware and ransomware.

Microsoft Defender for Endpoint is placed on the server side. It is an enterprise security solution that provides advanced threat protection, endpoint detection and response (EDR), and automated investigations. It secures critical server infrastructure against advanced attacks and unauthorized access.

For offsite data protection, Sync.com is utilized as the cloud storage encryption provider. Sync.com provides end-to-end encryption, which ensures that data being backed up and stored in the cloud is protected and private. This facilitates disaster recovery strategies and data protection compliance.

Backup and Configuration Management

For data integrity and system availability, backing-up procedures are automated through built-in Ubuntu backup tools and custom scripting programs. They govern scheduled backups, confirm data consistency, and facilitate rapid restoration when required. Windows and Ubuntu server systems are both supported by the same automated backup procedures, reducing the likelihood of data loss.

Both the Windows and the Ubuntu platforms are supported for configuration management through scripts and system software. The scripts and system software bridge the deployment of configuration files, upgrades, and system configuration, bringing consistency to the network and obviating manual mistakes. Configuration management through automation also makes it easy to scale and maintain, which can be crucial for running a WAN comprising numerous servers and services.

References

- Microsoft (2018) *Windows 10 Pro*. Available at: https://www.microsoft.com/en-us/windows/get-windows-10 (Accessed: 2 May 2025).
- Microsoft (2021) *Windows 11 Pro*. Available at: https://www.microsoft.com/en-us/windows/windows-11 (Accessed: 2 May 2025).
- Microsoft (2019) *Windows Server 2019*. Available at: https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2019 (Accessed: 2 May 2025).
- Canonical (2022) *Ubuntu Server 22.04 LTS*. Available at: https://ubuntu.com/download/server (Accessed: 2 May 2025).
- Oracle (2024) *MySQL 8.0 Database*. Available at: https://www.mysql.com/products/community (Accessed: 2 May 2025).
- PostgreSQL Global Development Group (2024) *PostgreSQL*. Available at: https://www.postgresql.org/download (Accessed: 2 May 2025).
- Malwarebytes (2024) *Malwarebytes for Windows*. Available at: https://www.malwarebytes.com (Accessed: 2 May 2025).
- Microsoft (2024) *Microsoft Defender for Endpoint*. Available at: https://www.microsoft.com/en-us/security/business/endpoint-security/microsoft-defender-endpoint (Accessed: 2 May 2025).
- **Microsoft** (2025) *Windows Server documentation*. Available at: https://learn.microsoft.com/en-us/windows-server/ (Accessed: 3 May 2025).
- **Canonical Ltd.** (2025) *Ubuntu Server documentation*. Available at: https://ubuntu.com/server/docs (Accessed: 3 May 2025).
- **Oracle Corporation** (2025) *MySQL Documentation*. Available at: https://dev.mysql.com/doc/ (Accessed: 3 May 2025).